

a Frame population

All examples, views, and frames



$$Q \in \mathbb{R}^{B \times 64}$$

Flatten without pooling;
gather across workers

b Center and scale

Per-dimension mean and scale

$$\mu_d = \frac{1}{B} \sum_i Q_{id}$$

$$s_d = \sqrt{\frac{1}{B} \sum_i (Q_{id} - \mu_d)^2} + \epsilon$$

$$\tilde{Q}_{id} = \frac{Q_{id} - \mu_d}{\text{sg}(s_d)}$$

$\epsilon = 10^{-6}$; sg: detached scale

c Random projections

256 unit directions
resampled each forward pass



$$w_k \sim \mathcal{N}(0, I)$$
$$w_k \leftarrow w_k / \|w_k\|_2$$
$$Y = \tilde{Q}W$$

d Quantile matching

Sort each projection across frames

$$Y_{(1)k} \leq \dots \leq Y_{(B)k}$$

$$g_i = \Phi^{-1}\left(\frac{i}{B+1}\right)$$

Match sorted values to
standard-normal quantiles

$$\mathcal{L}_{\text{center}} = \text{mean}_d \mu_d^2$$

$$\mathcal{L}_{\text{scale}} = \text{mean}_d (s_d - 1)^2$$

$$\mathcal{L}_{\text{shape}} = \text{mean}_{i,k} (Y_{(i)k} - g_i)^2$$